Written explanation - Back end work flow :-

- Why conversation history must be sent with every request

The AI model has no built-in-memory. It does not store or remember anything from previous requests. Every single API call is completely independent, the moment a response is sent back, the Ai forgets everything. It does not know my name, what i asked before, or what it just replied. It starts completely fresh each time. "To maintain context accross" multiple turns, the full conversation history must be included in every single request. Without it, our model would not know what was said earlier and would respond as if it were the very first message, making multi turn conversation impossible.

- How the system prompt changes between the three modes - give one example line for each mode.

When a user select a mode from the dropdown, the back-end reads that value and uses it to construct a different system prompt before making the AI API call. A system prompt is a hidden instruction that is sent to the AI model at the start of every request.

For Friendly mode, the Mock reply style is warm and welcoming for example "Hey there, I am happy to help".

For Professional Mode, the tone becomes more Formal and business-like for example " Hello, How may I assist you today?"

For concise mode the reply is intentionally shorter and more direct, For example: " Hi, what do you need? "

- How you would handle an API time out differently from a validation error.

An Api time out should be handled differently from a validation error because they are different problems. A Validation error means the request itself is wrong, so the backend should reject it immediately with a 400 response, For example " message must be 500 characters or fewer." An Api time out means the request was valid, but the AI provider did not respond in time, so the backend should return a 504 response. For example : " AI Service timed out"